

ABSTRACT

Table S1

HuggingFace Hub snapshot identifiers (SHA-1) for the three base-model checkpoints used in this study. The SHA-1 identifies the exact commit of the model repository that was loaded; weights can be obtained via `AutoModelForCausalLM.from_pretrained(repo_id, revision=sha)`.
†Retrieved without `revision=` parameter; inferred from HuggingFace cache active at execution time (see supplementary text above).

Backbone	Repository identifier	Snapshot SHA-1
Qwen 2.5-1.5B-Instruct	Qwen/Qwen2.5-1.5B-Instruct	989aa798 0e4cf806f80c7fef2b1adb7bc71aa306
Gemma 3 1B IT	google/gemma-3-1b-it	dcc83ea8 41ab6100d6b47a070329e1ba4cf78752
Gemma 4 E2B IT†	google/gemma-4-E2B-it	70af34e2 0bd4b7a91f0de6b22675850c43922a03

Table S1: Base-Model Checkpoint Identifiers

All three backbones were retrieved from the Hugging Face Hub without an explicit `revision=` argument; the loaded revision is therefore determined by the snapshot present in the local cache at the time of each experiment. The cache snapshot identifiers (commit SHA-1 hashes of the model repository at download time) are listed below and are sufficient to retrieve the identical weights from the Hub. The identifier for Gemma 4 E2B IT reflects the snapshot downloaded on 2026-06-26 and present in cache when the Gemma 4 experiments were executed (2026-06-27); a second snapshot (3e22461) was downloaded to the same cache on 2026-09-30 but was not used in any reported experiment.

The full 40-character SHA-1 strings are:

- Qwen 2.5-1.5B-Instruct: 989aa7980e4cf806f80c7fef2b1adb7bc71aa306
- Gemma 3 1B IT: dcc83ea841ab6100d6b47a070329e1ba4cf78752
- Gemma 4 E2B IT: 70af34e20bd4b7a91f0de6b22675850c43922a03

These identifiers are also listed in `configs/checkpoints.yaml` of the replication package (Zenodo DOI: 10.5281/zenodo.23146342).